The two first-order model operators are each other's adjoints when h has a positive lower bound

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Editorial scope. the counted statement and its own proof environment only

Allowed standard prerequisites (editorial).

- The two first-order model operators and their domains. The operators eq D0+ P and eq D0- 1-P acting on L^2(tilde S_C^+), with the positive function h on the cylinder.. The pair the proposition relates; declared as the article's standing notation, no external stub beyond the two operator-display stubs.

Source reference. Original equation 3.20: sources/sigma-2023-023-spectral-asymmetry-and-index-theory-on-manif/pinned-original-source.tex, source lines 1005–1005. Navigation pointer invoked at line(s) 1199: the counted text cites that numbered equation of the same article, which is not retained in this card, so the reference is declared as a pointer carrying the number the article itself prints for it (the section and the place of the equation inside that section, derived by the census v3.1 rule for the section-scoped equation counter); the cited equation is an allowed input and none of it is reproduced inside the span.

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Proposition 3.14. *If h has a positive lower bound, then the two operators* (3.20) *and* (3.21) *are each other's adjoints.*

Proof. We claim that for all $s_P \in W^1_{\widetilde{D}_C}\big(\widetilde{S}^+_C; P\big)$ and $s_{1-P} \in W^1_{\widetilde{D}_C}\big(\widetilde{S}^+_C; 1-P\big)$,

$$\big(\widetilde{D}^+_C s_P, s_{1-P}\big)_{L^2} = \big(s_P, \widetilde{D}^-_C s_{1-P}\big)_{L^2}. \tag{3.33}$$

Indeed, suppose that

$$s_P = s^N_P \otimes f_P, \qquad s_{1-P} = s^N_{1-P} \otimes f_{1-P},$$

for $s^N_P, s^N_{1-P} \in L^2\big(S^+|_N\big)$ and $f_P, f_{1-P} \in L^2[0,\infty)$ such that $s_P \in W^1_{\widetilde{D}_C}\big(\widetilde{S}^+_C; P\big)$ and $s_{1-P} \in W^1_{\widetilde{D}_C}\big(\widetilde{S}^+_C; 1-P\big)$. Then by self-adjointness of D^+_N and integration by parts, the left-hand side of (3.33) equals

$$\big(s^N_P, D^+_N s^N_{1-P}\big)_{L^2(S^+|_N)}\left(\big(f_P\overline{f_{1-P}}\big)|^\infty_0 + \int_0^\infty f_P(x)\big(-\overline{f'_{1-P}(x)} + h(x)\overline{f_{1-P}(x)}\big)\,\mathrm{d}x\right). \tag{3.34}$$

If we further decompose the expressions with respect to eigenspaces of D^+_N, and use that the components of f_P for positive eigenvalues equal zero at zero, and the components of f_{1-P} for negative eigenvalues equal zero at zero, then we find that the components for all eigenvalues of $f_P\overline{f_{1-P}}$ are zero at zero. So the term $\big(f_P\overline{f_{1-P}}\big)|^\infty_0$ in (3.34) vanishes, and (3.34) equals the right-hand side of (3.33).

Now if $\sigma_P := \widetilde{D}^+_C s_P$ and $\sigma_{1-P} := \widetilde{D}^-_C s_{1-P}$, then (3.33) becomes

$$\big(\sigma_P, \big(\widetilde{D}^-_C\big)^{-1}\sigma_{1-P}\big)_{L^2} = \big(\big(\widetilde{D}^+_C\big)^{-1}\sigma_P, \sigma_{1-P}\big)_{L^2}.$$

The sections σ_P and σ_{1-P} of this type are dense in $L^2\big(\widetilde{S}^+_C\big)$. And the inverse operators $\big(\widetilde{D}^+_C\big)^{-1}$ and $\big(\widetilde{D}^-_C\big)^{-1}$ are bounded, so we find that

$$\big(\big(\widetilde{D}^+_C\big)^{-1}\big)^* = \big(\widetilde{D}^-_C\big)^{-1}.$$

This implies that $\big(\widetilde{D}^+_C\big)^* = \widetilde{D}^-_C$. □